MTH404
Quiz#04
2 January 2025

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 1 of 10 (Start time: 08:00:56 AM, 02 January 2025) **Total Marks:** 1

If 10ft/s is the velocity of the projection, then the maximum horizontal range is .....

Select the correct option

Reload Math Equations

- ○ 10/32
- ● 100/32
- ○ 100*32
- ○ 10*32

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 2 of 10 (Start time: 08:01:45 AM, 02 January 2025) **Total Marks:** 1

If $45^0$and 20 are the angle and velocity of projection respectively, then the time of flight of the projectile is .....

Select the correct option

Reload Math Equations

- ○ 20g
- ○ $20\sqrt{2}$
- ○ $20\sqrt{2}g$
- ● $\frac{20\sqrt{2}}{g}$

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 3 of 10 (Start time: 08:02:27 AM, 02 January 2025) **Total Marks:** 1

Orbit of a particle moving under a central force is necessarily _____

Select the correct option

- (●) a plane curve
- ( ) none of these
- ( ) a straight line
- ( ) a space curve

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Question # 4 of 10 (Start time: 08:02:59 AM, 02 January 2025) **Total Marks:** 1

Areal speed of the radius vector r joining the particle to the centre of force is

Select the correct option

- (●) constant
- ( ) variable
- ( ) not constant
- ( ) none of these

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 5 of 10 (Start time: 08:03:29 AM, 02 January 2025) **Total Marks:** 1

If $45^0$and v_0 are the angle and velocity of projection respectively, then the time of flight of the projectile is .....

Select the correct option

Reload Math Equations

- ○ $v_0 * g$
- ○ $\sqrt{2}v_0$
- ○ $\sqrt{2}v_0 * g$
- ◉ $\frac{\sqrt{2}v_0}{g}$

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 6 of 10 (Start time: 08:04:10 AM, 02 January 2025) **Total Marks:** 1

The maximum horizontal range of a projectile is 80ft, the velocity of the projection is-----.

Select the correct option

Reload Math Equations

- ○ 45.5 ft/s
- ◉ 50.6 ft/s
- ○ 35 ft/s
- ○ 40.2 ft/s

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 7 of 10 (Start time: 08:04:48 AM, 02 January 2025) **Total Marks:** 1

The maximum horizontal range of a projectile is 100ft, the velocity of the projection is-----.

Select the correct option

- (●) 56.57ft/sec
- ( ) 3.6ft/sec
- ( ) 4.6ft/sec
- ( ) 5.6ft/sec

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 8 of 10 (Start time: 08:05:23 AM, 02 January 2025) **Total Marks:** 1

In a conservative field, the total energy of a particle through the motion

Select the correct option

- (●) remains constant
- ( ) varies rapidly
- ( ) does not remain constant
- ( ) none of these

MTH404 - Dynamics (Quiz 4) **Quiz Start Time: 08:00 AM, 02 January 2025**

Question # 9 of 10 (Start time: 08:05:59 AM, 02 January 2025) **Total Marks:** 1

The central force is ---------- if it is directed away from the centre.

Select the correct option

Reload Math Equations

- not attractive
- repulsive
- none of these
- attractive

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 08:06:34 AM, 02 January 2025) **Total Marks:** 1

If 45^0 and v_0 are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is .....

Select the correct option

Reload Math Equations

- $\frac{v_0}{\sqrt{2}g}$
- $v_0\sqrt{2}$
- $v_0\sqrt{2}g$
- $\frac{v_0}{\sqrt{2}}$